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Bicycle carrier with damper

Abstract

A bicycle carrier with a hitch mount includes a hinge assembly that couples a support post to a hitch bar configured for coupling to a vehicle hitch bar receiver, the hinge assembly including a damper to control a rate of rotation of the support post relative to the hitch bar. A plurality of wheel receiving members are coupled to a horizontal support member, which is in turn coupled to the support post. The hinge assembly may include one or more features configured to improve function of the bicycle carrier.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) The present application is a Continuation-in-Part of co-pending U.S. application Ser. No. 18/737,079, entitled “ROTATABLE HITCH-MOUNTED PAYLOAD ADAPTOR,” filed Jun. 7, 2024. U.S. application Ser. No. 18/737,079 claims priority benefit from U.S. Provisional Patent Application Ser. No. 63/626,355, entitled “ROTATABLE HITCH-MOUNTED PAYLOAD ADAPTOR,” filed Jan. 29, 2024. U.S. application Ser. No. 18/737,079, entitled “ROTATABLE HITCH-MOUNTED PAYLOAD ADAPTOR,” is also a Continuation-in-Part of U.S. application Ser. No. 18/155,696 filed Jan. 17, 2023, now U.S. Pat. No. 12,005,866, which issued on Jun. 11, 2024. U.S. patent application Ser. No. 18/155,696 is a Continuation-in-Part of U.S. application Ser. No. 17/339,562, entitled “BICYCLE RACK WITH C-HOOPS,” filed Jun. 4, 2021, which issued on Jan. 17, 2023 as U.S. Pat. No. 11,554,724. U.S. patent application Ser. No. 17/339,562 is a Continuation of U.S. application Ser. No. 17/003,094, entitled “BICYCLE CARRIER AND BICYCLE STORAGE RACK,” filed Aug. 26, 2020 which issued Feb. 7, 2023 as U.S. Pat. No. 11,572,022. U.S. application Ser. No. 17/003,094 is a Continuation-in-Part of then co-pending International Patent Application No. PCT/US2019/044865 entitled “BICYCLE CARRIER AND BICYCLE STORAGE RACK,” filed Aug. 2, 2019. International Patent Application No. PCT/US2019/044865 claims priority benefit from U.S. Provisional Patent Application No. 62/715,203, entitled “BICYCLE CARRIER AND BICYCLE STORAGE RACK,” filed Aug. 6, 2018, now expired. International Patent Application No. PCT/US2019/044865 also claims priority benefit from U.S. Provisional Patent Application No. 62/841,933, entitled “BICYCLE CARRIER AND BICYCLE STORAGE RACK,” filed May 2, 2019, now expired.

(1) Each of the foregoing applications, to the extent not inconsistent with the disclosure herein, is incorporated by reference in its entirety.

(2) This application is described by the extrajudicial term Continuation-in-Part to draw attention to the fact that its description differs from its immediate parent application. All terminology, structure, and function used in the description and claims is directly supported by one or more applications in its priority chain, each of which is incorporated by reference. Applicant asserts no new matter is added hereto.

SUMMARY

(3) In an embodiment, a bicycle carrier with a hitch mount for a vehicle includes a hitch bar configured to mount to a vehicle trailer hitch receiver; a hinge assembly operatively coupled to the hitch bar, the hinge assembly including a damper operatively coupled to a support post and the hitch bar, the damper being configured to limit a rate of rotation of the support post relative to the hitch bar. The support post has a proximal end and a distal end, the support post being operatively coupled to the hinge assembly at the proximal end, the support post being rotatable relative to the hitch bar through an arc in a vertical plane defined by the hinge assembly between a lowered position and a raised position. A first support member is operatively coupled to the distal end of the support post. A wheel-receiving member is coupled to the first support member, the wheel-receiving member being configured to receive a wheel of a bicycle such that the bicycle is suspended vertically while the wheel is in the wheel-receiving member, the wheel-receiving member being further configured to receive the wheel at any of a range of angles with respect to the support post, the range of angles including at least an angle substantially parallel to the support post and an acute angle with respect to the support post.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is a perspective view of a bicycle carrier with a hitch mount, according to an embodiment.
- (2) FIG. 2 is an illustration of a bicycle carrier coupled to a vehicle and carrying a bicycle, according to an embodiment.
- (3) FIG. 3 illustrates a bicycle carrier carrying a bicycle and coupled to a vehicle while at a loading position, according to one embodiment.
- (4) FIG. 4 is a side view of a bicycle carrier coupled to a vehicle and in a stabilization configuration, according to an embodiment.
- (5) FIG. 5 is a side view of a bicycle carrier positioned on the ground and stabilizing a bicycle, according to an embodiment.
- (6) FIG. 6 is a view of the bicycle carrier during a loading sequence of the bicycle, according to an embodiment.
- (7) FIG. 7 is a view of the bicycle carrier after a loading sequence of the bicycle, according to an embodiment.
- (8) FIG. 8 is a view of a bicycle carrier supporting a first bicycle and a second bicycle mounted in a tune-up position, according to an embodiment.
- (9) FIG. 9A is a side view of a bicycle carrier in a wall-mounted configuration, according to an embodiment.
- (10) FIG. 9B is a view of a wall mount system used in the wall-mounted configuration of the bicycle carrier described in FIG. 9A, according to an embodiment.
- (11) FIG. 9C is a detailed view of the wall mount fittings shown in FIG. 9B, according to an embodiment.
- (12) FIG. 10 is a side view of a bicycle carrier in a loading position, according to an embodiment.
- (13) FIG. 11 is a side view of a bicycle carrier in a storage position, according to an embodiment.
- (14) FIG. 12 is a perspective view of a wheel-receiving member formed as a C-hoop and angled support members, according to an embodiment.
- (15) FIG. 13 is a perspective view of a C-hoop and angled support members, according to an alternate embodiment.
- (16) FIG. 14 is an isometric view of a hinge assembly of a bicycle carrier and wedge, according to an embodiment.

- (17) FIG. 15 illustrates four views of a hinge assembly of a bicycle carrier, according to an embodiment.
- (18) FIG. 16 is a bottom sectional view of the hinge assembly and a hitch bar, according to an embodiment.
- (19) FIG. 17 illustrates two views of a hinge assembly of a bicycle carrier, according to an embodiment.
- (20) FIG. 18 is a cross section view of a bicycle carrier post showing the lever actuation of the locking latch pin, according to an embodiment.
- (21) FIG. 19 is a side sectional view of a first horizontal support member of a bicycle carrier showing a stored internal locking cable, according to an embodiment.
- (22) FIG. 20 illustrates a bicycle carrier that can telescope to reduce the number of C-hoops, according to an embodiment.
- (23) FIG. 21 illustrates a C-hoop and a portion of angled support members, according to an embodiment.
- (24) FIG. 22A is a diagram showing a portion of a sports equipment carrier including a force transmission in a raised position, according to an embodiment.
- (25) FIG. 22B is a diagram showing a portion of a sports equipment carrier including the force transmission of FIG. 22A in a lowered or partially lowered position, according to an embodiment.
- (26) FIG. 23 illustrates a sports equipment carrier configured to carry a combination of bicycles and whitewater kayaks, according to an embodiment.

DETAILED DESCRIPTION

(27) In the following detailed description, reference is made to the accompanying drawings, which form a part hereof. In the drawings, similar symbols typically identify similar components, unless context dictates otherwise. Other embodiments may be used and/or other changes may be made without departing from the spirit or scope of the disclosure.

(28) FIG. 1 is a perspective view of a bicycle carrier **100**, according to an embodiment. As used herein, the terms bicycle carrier, sports equipment carrier, sports rack, and the like are used interchangeably. Unless context dictates otherwise, the terms should be considered synonymous. The bicycle carrier **100** includes a post **102** (also referenced as “support post” herein), an equipment mount **103**, a hitch bar **112**, and a hinge assembly **114** that rotatably couples the hitch bar **112** to the post **102**. In an embodiment, the equipment mount **103** includes a first horizontal support member **104**, wheel receiving members **106** including angled support members **108** configured to receive and hold bicycle wheels, and a second horizontal support member **110**. The post **102** and/or equipment mount **103** may optionally include a cable lock loop **115** configured to receive one or more locking cables or chains for securing bicycles to the bicycle carrier **100**.

(29) In the view of FIG. 1, the bicycle carrier **100** is in a raised position for transport. As used herein, the terms raised position and transport position will be understood as synonymous unless context dictates otherwise. In the transport position the hinge assembly **114** holds the post **102** at a position substantially perpendicular to the hitch bar **112**. The hitch bar **112** is configured to be placed in and fixed to a trailer hitch receiver of a vehicle. In the transport position, the bicycle carrier **100** can safely carry a plurality of bicycles while coupled to a moving vehicle **124**.

(30) Each wheel receiving member **106** is configured to receive a front wheel of a bicycle. The front wheel of a bicycle can be positioned, lowered into, or rolled into, a wheel receiving member **106**. After being positioned in a wheel receiving member **106**, the wheel of the bicycle comes to rest at an angle other than vertical. The wheel of the bicycle rests against the angled support members **108** and on the first horizontal support member **104**. The bicycle is suspended vertically when the wheel is positioned in the wheel receiving member **106**. This is illustrated in FIG. 2 and elsewhere in the figures. The angled support members **108** can be coupled to the first horizontal support member **104** by bolts, welds, or other coupling devices or methods.

(31) As used herein, it will be understood that a wheel receiving member may take the form of a C-

shaped hoop, also referred to as C-hoop. In description herein, unless context dictates otherwise, it will be understood that the term C-hoop may be generalized to wheel receiving member. This naming convention is supported by one or more patents and applications from which this application derives priority and incorporates by reference.

(32) Each wheel receiving member **106** is coupled to the first horizontal support member **104** by one or more angled support members **108**. In the example of FIG. 1, the support members **108** extend from the first horizontal support member **104** at an angle other than vertical. In one embodiment, the support members **108** extend at an angle of about 55° from horizontal. In this case, the wheel of the bicycle will rest in a wheel receiving member **106** at an angle of about 45° from horizontal. Those of skill in the art will recognize, in light of the present disclosure, that other angles are possible without departing from the scope of the present disclosure.

(33) In one embodiment, the post **102** is about 48 inches in length. The post **102** can have width dimensions of about 2 inches by 3 inches. The first horizontal support member **104** is about 68 inches in length. The first horizontal support member **104** has width dimensions of about 2 inches by 2 inches. The second horizontal support member **110** has a length of about 64 inches. The second horizontal support member **110** has width dimensions of about 1.75 inches by 1.75 inches. The hitch bar **112** has a length of about 20 inches, with width dimensions of about 2 inches by 2 inches. In one embodiment, the post **102**, the first horizontal support member **104**, the second horizontal support member **110**, and the hitch bar **112** are each made of steel. Those of skill in the art will recognize, in light of the present disclosure, that dimensions and materials described herein are given by way of example and that other suitable dimensions and materials can be utilized without departing from the scope of the present disclosure.

(34) In one embodiment, the first horizontal support member **104** is coupled to the post **102** by mounting tabs **118**. The mounting tabs **118** can include steel and can be fixed to the first horizontal support member **104** by bolts, by welding, or by other fasteners or methods. The mounting tabs **118** can be fixed to the post **102** by welding, bolts, or by other fasteners or methods.

(35) In one embodiment, the post **102** includes a release lever **116**. Pulling the release lever **116** enables the post **102** to be rotated relative to the hitch bar **112** via the hinge assembly **114**. The post **102** can be rotated between various stopping positions, as will be described in greater detail below. The structure, components, and function of the hinge assembly **114** will also be described in greater detail below with respect to FIGS. 14-17.

(36) In one embodiment, the bicycle carrier **100** includes a plurality of first straps **120** each coupled to the first horizontal support member **104**. There is a first strap **120** for each wheel receiving member **106**. When a bicycle wheel is placed in a wheel receiving member **106**, the corresponding first strap **120** is looped around the wheel between two of the spokes and is fastened to a strap anchor on one of the support members **108**. This can keep the wheel from moving or shifting when positioned in the wheel receiving member **106**.

(37) In one embodiment, the bicycle carrier **100** includes a plurality of second straps **122** each coupled to the second horizontal support member **110**. There is a second strap **122** for each wheel receiving member **106**. When a bicycle is suspended from a wheel receiving member **106** by placing the wheel in the wheel receiving member **106**, the bicycle will be suspended vertically, and the rear wheel will be in contact with the second horizontal support member **110**. The second strap **122** can be looped between two spokes on the rear wheel and connected to a fastener positioned on the second horizontal support member **110**. This will keep the rear wheel from shifting relative to the second horizontal support member **110** during transport. Those of skill in the art will recognize, in light of the present disclosure, that other fastening methods can be used to secure the bicycle suspended from one of the wheel receiving members **106**.

(38) The word vertical in the term post **102** refers to the position of the post **102** when in the transport configuration. The transport configuration corresponds to the configuration in which the bicycle carrier **100** is coupled to a vehicle, is carrying one or more bicycles **126**, **126a**, **126b**, and is

actively being transported. In this case, the post **102** is substantially perpendicular to the hitch bar **112** and extends substantially vertically relative to flat ground. Those of skill in the art will recognize, that the post **102** can be rotated to positions other than vertical. Additionally, the bicycle carrier **100** can be laid in positions in which the post **102** would not extend in a vertical direction. Accordingly, the word vertical in the term post **102** refers to the orientation of the post **102** in a particular transport configuration. Similarly, the word horizontal in the terms first and second horizontal support member **104**, **110**, refers to the orientation of the support member **104**, **110** during the transport configuration. The bicycle carrier **100** could be laid in positions in which the first and the second horizontal support members **104**, **110** would not extend in the horizontal direction. In one embodiment, the bicycle carrier **100** can include a post **102** that is not necessarily vertical in the transporting position. Additionally, the bicycle carrier **100** can include first and second support members **104**, **110** that are not horizontal.

(39) FIG. 2 is an illustration of a bicycle carrier **100** coupled to a vehicle **124** and carrying a bicycle **126**, according to an embodiment. The hitch bar **112** of the bicycle carrier **100** has been inserted in and coupled to a trailer hitch receiver **127** of the vehicle **124**. The hitch bar **112** can be coupled to the trailer hitch receiver **127** by bolts and pins or other common fastening configurations, as will be described in more detail below.

(40) The front wheel **128** of the bicycle **126** is positioned in one of the wheel receiving members **106**, here depicted as a C-hoop. When the front wheel **128** is positioned in the C-hoop **106**, the front wheel **128** rests at a non-vertical angle. The front wheel **128** rests on the first horizontal support member **104**. The front wheel **128** is also in contact with one or more support members **108**. The bicycle **126** is suspended vertically. The rear wheel **129** rests against the second horizontal support member **110**. When describing how the front and the rear wheels **128**, **129** rest on or are supported by various support members, it is understood that the tires may actually be in contact with the various support members.

(41) The post **102** is in the transport position in which the post **102** extends substantially perpendicularly from the hitch bar **112**.

(42) While FIG. 2 shows a single bicycle **126** carried by the bicycle carrier **100**, in practice, the bicycle carrier **100** can carry multiple bicycles **126**. In particular, the bicycle carrier **100** can carry a bicycle **126** for each C-hoop **106**. Accordingly, each C-hoop **106** can receive a front wheel **128** of a respective bicycle **126**.

(43) FIG. 3 illustrates a bicycle carrier **100** carrying a bicycle **126** and coupled to a vehicle **124** while at a loading position, according to one embodiment. The hinge assembly **114** defines a loading stop position. When the release lever **116** is pulled, the hinge assembly **114** enables the post **102** to rotate relative to the hinge bar **112** to a loading stop position. At the loading stop position, the post **102** is oriented at about 33° from vertical, in one example. This enables bicycles **126** to more easily be loaded onto or off of the bicycle carrier **100**. This is because the C-hoops **106** are both lower to the ground and oriented at an angle that facilitates easy loading and unloading of bicycles **126**.

(44) As will be described in more detail below, hinge motion is slowed by dampers/gas springs included in the hinge assembly **114**. This damping enhances the safety of rotating between the transport position and the load position.

(45) In one embodiment, each bicycle **126** can be accessed without the removal of other bicycles **126**. Loading bicycles **126** can also be done in the full vertical transport mode or in the angled back loading position. In one embodiment, moving from the hinged position requires an upward and forward force toward the vehicle **124** to move from the loading (hinged) position until the post **102** automatically locks in the vertical or transport position.

(46) FIG. 4 is a side view of a bicycle carrier **100** coupled to a vehicle **124** and in a stabilization configuration, according to an embodiment. In the stabilization configuration, the post **102** is rotated more than 90° from the transport position by operation of the hinge assembly **114**. The C-

hoops **106** rest on the ground. A bicycle **126** is stabilized by the bicycle carrier **100**. In particular, the rear wheel **129** of the bicycle **126** is positioned in the C-hoop **106**. The C-hoop **106** stabilizes the rear wheel **129** so that the bicycle **126** will not fall over. The first strap **120** can be utilized to strap the rear wheel **129** to the C-hoop **106**.

(47) This configuration is useful in situations in which the vehicle **124** will be parked and bicycles **126** will be periodically used. When a bicycle **126** is not used, it can be positioned in the C-hoop **106** as shown in FIG. 4. When a bicycle **126** is to be used, the bicycle **126** can easily be removed from the C-hoop **106**. This configuration can be particularly useful in situations like camping when bicycles **126** may be ridden to and from camp. Instead of being laid on the ground and are rested against trees or other structures, each bicycle **126** can be positioned in a C-hoop **106** for stabilization.

(48) FIG. 5 is a side view of a bicycle carrier **100** positioned on the ground and stabilizing a bicycle **126**, according to an embodiment. In this example, the post **102** and the hinge part **112** are oriented in the transport position by the hinge assembly **114**. However, the bicycle carrier **100** is not coupled to a vehicle **124**. Instead, the bicycle carrier **100** is laid on the ground. The C-hoops **106** rest on the ground similar to the configuration shown in FIG. 4. The bicycle carrier **100** can stabilize a plurality of bicycles **126** in this configuration by positioning the rear wheels **129** of the bicycles **126** in the C-hoops **106**.

(49) FIG. 6 is a view of the bicycle carrier **100** during a loading sequence of the bicycle **126**, according to an embodiment. The front wheel **128** of the bicycle **126** is lowered into the C-hoop **106**. While the front wheel **128** is loaded into the C-hoop **106**, the front wheel **128** can be substantially vertically oriented. After loading, the front wheel **128** will settle at an angle other than vertical.

(50) FIG. 7 is a view of the bicycle carrier **100** after a loading sequence of the bicycle **126**, according to an embodiment. The front wheel **128** of the bicycle **126** is positioned in the C-hoop **106**. The front wheel **128** has settled to a position such that the front wheel **128** is not oriented vertically, but rather is oriented at an angle relative to vertical. In one example, the angle is 45°.

(51) FIG. 8 is a view of a bicycle carrier **100** supporting a first bicycle **126a** and a second bicycle **126b** mounted in a tune-up position, according to an embodiment. The first bicycle **126a** is supported in a standard fashion with the front wheel **128** positioned in a C-hoop **106**. The second bicycle **126b** is supported by positioning a seat post within a C-hoop **106** such that a seat **125** of the second bicycle **126b** rests on and is supported by the C-hoops **106**.

(52) The shape of the C-hoop **106** enables the second bicycle **126b** to be suspended by the seat **125**. Because the C-hoop **106** includes first and second terminations **130**, **132** (see FIGS. 12 and 13), the seat post of the second bicycle **126b** can be easily maneuvered through a gap **131** (see FIGS. 12 and 13). The second bicycle **126b** can then be suspended by the seat **125**, as shown in FIG. 8.

(53) In one embodiment, when the second bicycle **126b** is suspended by the seat **125**, maintenance or repairs can be performed easily on the second bicycle **126b** because the wheel and pedals are free to rotate. Accordingly, the bicycle carrier **100** can act as a repair station for bicycles **126**.

(54) FIG. 9A is a side view of a bicycle carrier **100** in a wall-mounted configuration, according to an embodiment. The hinge assembly **114** is operated such that the post **102** has rotated more than 90° from the transport position to a storage or wall-mounted position. In one example, in the wall-mounted or storage position, the post **102** is rotated about 150° relative to the hinge bar **112** from the transport position.

(55) In one embodiment, the wall-mounted position is suitable for coupling the bicycle carrier **100** to a wall mount structure **144**. The wall mount structure **144** can include one or more boards, beams, or bent sheet metal coupled to a wall **145** of a structure, such as a garage. One or more hooks **146** are coupled to the wall mount structure **144**. The bicycle carrier **100** is coupled to the wall mount structure **144** by positioning one or more of the C-hoops **106** on one or more of the hooks **146**. The post **102** and the hitch bar **112** are in contact with a floor **148** or on top of a small

spacer sitting on the floor **148**.

(56) In the wall-mounted position, a bicycle **126** can be loaded into a C-hoop **106** by rolling the back wheel **129** with the front wheel **128** in the air until the front wheel **128** can be positioned in the C-hoop **106**. In some cases, this may require lifting the bicycle **126** to enable the front wheel **128** to slide into the C-hoop **106**. The downward tilting angle of the C-hoop **106** makes it relatively easy to place the front wheel **128** of a bicycle **126** in the C-hoop **106**. Each C-hoop **106** of the bicycle carrier **100** can receive a respective bicycle **126** in the wall-mounted configuration.

(57) FIG. **9B** is a view of a wall mount system used in the wall-mounted configuration of the bicycle carrier **100** described in FIG. **9A**, according to an embodiment. FIG. **9C** is a detailed view of the wall mount fittings **149** described in FIG. **9B**, according to an embodiment.

(58) Referring to FIGS. **9B** and **9C**, in an embodiment, the bicycle carrier **100** further includes a wall mount bar **144** (e.g., wall mount structure **144** of FIG. **9A**) configured to be screwed to a wall **145**, such as using wood screws to be screwed into wall studs, and one or more wall mount fittings **149** (such as one or more hooks **146**) configured to be coupled to the wall mount bar **144** and configured to receive and hold a feature of the equipment mount **103**. In an embodiment, the wall mount system uses two wall mount fittings **149**, the two wall mount fittings **149** being configured to hook around respective C-hoops **106**.

(59) FIG. **10** is a side view of a bicycle carrier **100** in a loading position, according to an embodiment. The hinge assembly **114** has enabled the post **102** to rotate about 33 degrees from the transport position to the loading position.

(60) FIG. **11** is a side view of a bicycle carrier **100** in a storage position, according to an embodiment. The hinge assembly **114** has enabled the post **102** to rotate about 150 degrees from the transport position to the storage position.

(61) FIG. **12** is a perspective view of a C-hoop **106** and angled support members **108**, according to an embodiment. The C-hoop **106** is substantially in the shape of a letter C. The C-hoop **106** includes two terminations, **130**, **132**. The two terminations **130**, **132** define a gap **131** between the two terminations **130**, **132**. The C-hoop **106** includes a first bend **134**, a second bend **136**, and a straight portion **138** extending between the first and the second bends **134**, **136**.

(62) In one embodiment, the length of the C-hoop **106** is 26 inches. The length is defined as the internal distance between the farthest points of the first bend **134** and the second bend **136**. Those of skill in the art will recognize, in light of the present disclosure, that a C-hoop **106** can have a length other than 26 inches without departing from the scope of the present disclosure.

(63) In one embodiment, the width of the C-hoop **106** is 3.5 inches. The width is defined as the shortest distance between the straight portion **138** and either of the terminations **130** or **132**. Those of skill in the art will recognize, in light of the present disclosure, that a C-hoop **106** can have a width other than 3.5 inches without departing from the scope of the present disclosure.

(64) In one embodiment, the distance D between the first termination **130** and the second termination **132** is about 10 inches. Accordingly, the gap **131** between the first and the second terminations **130**, **132** is 10 inches. Those of skill in the art will recognize, in light of the present disclosure, that the C-hoop **106** can include a gap **131** of other than 10 inches between the first and the second terminations **130**, **132** without departing from the scope of the present disclosure.

(65) In one embodiment, the shape of the C-hoop **106** provides many benefits. For example, when a front wheel **128** is inserted into the C-hoop **106**, the front wheel **128** can enter at an angle substantially perpendicular to the plane defined by the C-hoop **106**. Because the C-hoop **106** terminates at the terminations **130**, **132** without additional structure extending downward from the area of the terminations **130**, **132**, the front wheel **128** can be loaded at a convenient perpendicular angle. If the C-hoop **106** did not terminate at the terminations **130**, **132**, but rather extended downward at an angle similar to the angled support members **108**, then the front wheel **128** would need to be loaded at an angle other than perpendicular so it can slide into the C-hoop **106**. Loading the front wheel **128** at an angle other than perpendicular is highly inconvenient and awkward.

Accordingly, the shape of the C-hoop **106** as shown in FIG. **12** avoids this drawback. After the front wheel **128** has been inserted into the C-hoop **106**, the front wheel **128** will then come to a resting position at an angle other than perpendicular. In the resting position, the tire is in contact with an interior surface of the angled support members **108** and the first horizontal support member **104**.

(66) Another benefit of the C-hoop **106** is that the gap **131** is situated so that in the resting position, the spokes of the front wheel **128** do not contact the C-hoop **106**. If the C-hoop **106** was a complete hoop instead of having the gap **131**, then it is possible that the spokes would be in contact with the C-hoop **106** when in the resting position. The resulting pressure on the spokes could result in bent or otherwise damaged spokes.

(67) In one embodiment, the C-hoop **106** is a single integral structure. For example, the C-hoop **106** can correspond to a single bar or rod that is bent into the shape shown in FIG. **12**.

Alternatively, the C-hoop **106** can include multiple sections connected together to form the shape shown in FIG. **12**.

(68) In one embodiment, the C-hoop **106** is steel. Other sufficiently strong materials can be used for the C-hoop **106**.

(69) In one embodiment, when the bicycle carrier **100** is in the transport position, the C-hoop **106** defines a plane substantially parallel to the ground, or perpendicular to the vertical direction. For example, if a flat board is laid on top of the C-hoop **106** when in the transporting configuration, the plane of the board would be parallel to the ground or perpendicular to the vertical direction.

(70) FIG. **13** is a perspective view of a C-hoop **106** and angled support members **108**, according to an alternate embodiment. The C-hoop **106** is substantially in the shape of a C. However, instead of a straight portion **138** extending between the first and the second bends **134**, **136**, a bent portion **142** extends between the first and the second bends **134**, **136**. Otherwise, the C-hoop **106** of FIG. **13** is substantially similar to and provides similar benefits as the C-hoop **106** of FIG. **12**.

(71) FIGS. **14-16** include illustrations of a hitch bar wedge clamp, according to an embodiment.

(72) FIG. **14** is an isometric view of a hinge assembly **114** of a bicycle carrier **100** and wedge **160**, according to an embodiment. FIG. **14** also shows portions of the hitch bar **112** and post **102** coupled to the hinge assembly **114**. The hinge assembly **114** allows rotation of the post **102** relative to the hitch bar **112**.

(73) In one embodiment, the hinge assembly **114** includes a bracket **150**, dampers **152**, a damper yoke **154**, a damper pin **155**, a hinge stop pin **156**, and a bolt **159**. The bracket **150** is fixed to the hitch bar **112** by one or more bolts or welds. The bolt **159** is coupled to the bracket **150** and passes through the post **102**. The bolt **159** acts as an axle for rotation of the post **102**. When the post **102** rotates, it rotates about the bolt **159**.

(74) In one embodiment, the bracket **150** is a unitary structure. In one embodiment, the bracket **150** is steel.

(75) In one embodiment, when the release lever **116** is pulled, the hinge assembly **114** enables rotation of the post **102** relative to the hitch bar **112**. The dampers **152** slow, resist, or oppose the rotation. This enhances the safety of the bicycle carrier **100**. This is because when rotating between the transport and the loading positions, the rotation cannot happen rapidly due to the dampers **152**. This ensures that the post **102** will not quickly fall or rotate when rotation is enabled, potentially injuring somebody or damaging the bicycle carrier **100**. Instead, after rotation is enabled by pulling the release lever **116**, some force may need to be applied by an individual to cause rotation against the force of the dampers **152** when no bicycles are on the rack.

(76) In one embodiment, the dampers **152** include hydraulic damping. The dampers **152** can include the fluid chamber and a piston. Flow of a fluid into or out of the fluid chamber is inhibited by a relatively small aperture, and potentially by the viscosity of the fluid. Rotation of the post **102** relative to the hitch bar **112** is opposed by this hydraulic effect. In one embodiment, the fluid is a liquid. In one embodiment, the fluid is an oil. In one embodiment, the fluid is a gas.

(77) In one embodiment, the dampers **152** can include one or more springs that resist rotation of the post **102** relative to the hitch bar **112**. In one embodiment, the dampers **152** can include gas springs. Those of skill in the art will recognize, in light of the present disclosure, that many other types of dampers **152** can be used without departing from the scope of the present disclosure.

(78) In one embodiment, the dampers **152** are mounted between the hitch bar **112** and the damper yoke **154**. The dampers **152** are coupled to the hitch bar **112** by a bolt **161**.

(79) In one embodiment, the hinge stop pin **156** defines a stop point for rotation of the post **102** relative to the hitch bar **112**. In particular, when rotating from the transport position to the loading position, the hinge stop pin **156** stops rotation at the transport position. In one embodiment, the loading position is a 33° rotation from the transport position. The bracket **150** includes multiple sets of apertures **176**, **177**, **178** for receiving the hinge stop pin **156**. The hinge stop pin **156** is placed in the apertures **177** in FIG. **14**. The hinge stop pin **156** can be placed in the apertures **178** to reduce the angle of the loading position relative to the transport position. The hinge stop pin **156** can be placed in the apertures **176** to substantially prohibit rotation of the post **102** from the transport position. The hinge stop pin **156** can be tied to the bracket **150** by a tie.

(80) The post includes **102** include first damper pin receivers **157** for receiving a damper pin **155**. The damper yoke **154** is placed in the damping position as shown in FIG. **14**. In the damping position, the dampers **152** dampen or resist rotation of the post **102**. The damper pin **155** is placed in the first damper pin receivers **157**, thereby fixing the damper yoke **154** to the post **102** in the damping position. The damper pin **155** can be tied to the damper yoke **154** by a tie.

(81) In one embodiment, to enable rotation of the post **102** to the wall mounting position, the damper pin **155** and the hinge stop pin **156** are removed. With the damper pin **155** removed, the damper yoke **154** can be decoupled from the post **102** and moved to a damper release position in which the dampers **152** do not dampen or resist rotation of the post **102**. With the damper pin **155** removed from the first damper pin receivers **157**, the hinge stop pin **156** removed, and the release lever **116** pulled, the post **102** can rotate beyond the loading position to the wall mounting position. This can correspond to rotation of up to 150 degrees from the transport position.

(82) The hitch bar **112** includes a second damper pin receiver **158**. The damper pin **155** can be received in the second damper pin receiver **158** to secure the damper yoke **154** in the damper release position.

(83) In one embodiment, a hitch wedge **160** helps to secure the hitch bar **112** in the hitch receiver **127** (e.g., trailer hitch receiver **127**) of a vehicle **124**. For example, after the hitch bar **112** is positioned in the hitch receiver **127**, the hitch wedge **160** will be positioned within the hitch receiver **127**. A hitch pin **162** can then be placed through apertures in the hitch receiver **127** and apertures **170** (see FIG. **15**) in the hitch bar **112**. A hitch pin lock **172** can lock the hitch pin **162** in place. A hitch wedge bolt **166** (see FIG. **15**) can then be tightened. The tightening of the hitch wedge bolt **166** pulls or tightens the hitch wedge **160** into place, thus locking the entire bicycle carrier **100** to the vehicle hitch mounting point and stopping relative motion between the bicycle carrier **100** and the vehicle **124**. This greatly reduces rattling of the bicycle carrier **100** and, by extension, the bicycles **126**, during transport. Loosening the hitch wedge bolt **166** loosens the hitch wedge **160**, allowing a hitch wedge spring **164** (see FIG. **15**) to fully release the hitch wedge **160**, thereby allowing the bicycle carrier **100** to be removed from the vehicle **124**. FIG. **14** also illustrates an expanded view of the hitch wedge **160**.

(84) In one embodiment, as the hitch wedge **160** is tightened, the hitch wedge **160** presses against an interior of the hitch receiver **127** of the vehicle **124**, preventing motion of the hitch bar **112** relative to the hitch receiver **127** of the vehicle **124**. This effect can be sufficient to secure the bicycle carrier **100** to the vehicle **124** without using the hitch pin **162**.

(85) In one embodiment, the hinge assembly **114** also includes a damper pin **155**. By pulling the damper pin **155**, the dampers **152** are decoupled from the post **102**. The damper yoke **154** and the dampers **152** can be rotated downward and the damper pin **155** can be placed in the second damper

pin receiver **158**. In this configuration, the post **102** can rotate beyond the loading position into the wall mounting or storage position.

(86) FIG. **15** illustrates four views **1500a**, **1500b**, **1500c**, **1500d**, of a hinge assembly **114** of a bicycle carrier **100**, according to an embodiment. View **1500a** is a side view of a portion of the hinge assembly **114**, the hitch bar **112**, and the portion of the post **102** in the transport position. View **1500b** is a side sectional view of the hinge assembly **114**, the hitch bar **112**, and a portion of the post **102** in the transport position. View **1500c** is a side view of the hinge assembly **114**, the hitch bar **112**, and a portion of the post **102** in the loading position. View **1500d** is a side sectional view of the hinge assembly **114**, the hitch bar **112**, and a portion of the post **102** in the loading position.

(87) In view **1500a**, the post **102** is in the transport position. The post **102** is substantially perpendicular to the hitch bar **112**. The dampers **152** are in a retracted position.

(88) In view **1500b**, the post **102** is in the transport position. A release rod **174** extends through the post **102**. The release rod **174** can be a release cable. When the release lever **116** is pulled, a locking latch pin **168** is released by action of the release rod **174**. This enables rotation of the post **102**. The side sectional view of **1500b** also illustrates the hitch wedge bolt **166** extending through the hitch bar **112** to the hitch wedge **160**. The side sectional view of **1500b** also illustrates apertures **170** in the hitch bar **112**, through which the hitch pin **162** can be passed.

(89) In view **1500c**, the post **102** is in the loading position. The locking latch pin **168** has been released, enabling rotation of the post **102** to the loading position. The loading position is achieved when the post **102** abuts the hinge stop pin **156**. The hinge stop pin **156** can be placed in the aperture **178** (see view **1500b**) to reduce the rotation angle of the post **102** relative to vertical. If the hinge stop pin **156** is placed in the aperture **176** (see view **1500b**), rotation of the post **102** from the transport position is prohibited. This locks the post **102** in the transport position. In the view **1500d**, the interior of the post **102** can be seen at the loading position.

(90) FIG. **16** is a bottom sectional view of the hinge assembly **114** and the hitch bar **112**, including a wedge **160**, according to an embodiment. From the bottom sectional view, a wedge spring bracket **180**, the hitch wedge spring **164**, and a wedge spring stop **182** are apparent.

(91) In one embodiment, the hitch bar **112** extends full length, greatly simplifying the insertion process because the load of the rack can be fully transferred to the hitch bar **112** as it is inserted into the hitch receiver **127**. The hitch bar **112** may be formed with one upper corner and about $\frac{1}{4}$ of the top wall and side wall abutting the upper corner extending the full length. The wedge **160** is formed to nominally occupy a space sloping away from the one upper corner, such that tightening the hitch wedge bolt **166** causes the wedge to slide along the sloping end of the hitch bar **112** to expand and securely hold the hitch bar in place. The internal spring **164** substantially prevents the wedge **160** from engaging too early in the insertion process, allowing it to be inserted easily. The orientation of the wedge spring bracket **180**, wedge spring stop **182** and spring **164** in the wedge **160** also provides pressure on the wedge **160** to hold it in place that keeps it lined up with the hitch bar **112** for easier insertion.

(92) FIG. **17** illustrates two views **1700a** and **1700b** of a hinge assembly **114** of a bicycle carrier **100**, according to an embodiment. View **1700a** is a side view of the hinge assembly **114**, the hitch bar **112**, and a portion of the post **102** in the wall mounting or storage position. View **1700b** is a side sectional view of the hinge assembly **114**, the hitch bar **112** and a portion of the post **102** in the transport position. To enter into the storage or the mounting position, the damper pin **155** is removed, the damper yoke **154** is removed and lowered to the hitch bar **112** and the damper pin **155** is placed in the second damper pin receiver **158**. With the dampers **152** and the damper yoke **154** decoupled from the post **102**, and the hinge stop pin **156** removed, the post **102** can rotate beyond the loading position into the storage or wall mounting position. In the storage or wall mounting position, the post **102** is rotated more than 90 degrees from the transport position, for example 150 degrees.

(93) FIG. 18 is a cross section view of a bicycle carrier post **102** showing the lever actuation of the locking latch pin **168**. The rack release lever **116** is moved downward causing the locking latch pin **168** to release the post **102** to rotate to one of the stopping positions, such as the transport position, the loading position, or the mounting or storage position.

(94) FIG. 19 is a side sectional view of a first horizontal support member **104** of a bicycle carrier **100**, according to an embodiment. The view of FIG. 19 illustrates that the first horizontal support member **104** is hollow. An internal locking cable **184** is positioned within the hollow first horizontal support member **104**. An end of the first horizontal support member **104** can be opened and the internal locking cable **184** can be drawn out. The internal locking cable **184** can then be used to lock any or all of the bicycles **126** that are carried by the bicycle carrier **100**. FIG. 19 also illustrates a strap anchor **186** coupled to the support member **108**. The first strap **120** can be looped between the spokes of a front wheel **128** and then latched onto the strap anchor **186**.

(95) FIG. 20 illustrates a bicycle carrier **100** that can telescope to reduce the number of C-hoops **106**, according to an embodiment. The first and the second horizontal support members **104**, **110** can telescope between maximum lengths and minimum lengths. For example, after removing the two C-hoops **106** on either end of the first horizontal support member **104** via removing screws or bolts, portions of the first horizontal support member **104** can be removed from or telescoped into an interior of the first horizontal support member **104**. The second horizontal support member **110** can be removed or telescoped in the same manner. In one example, the bicycle carrier **100** includes a maximum number of seven C-hoops **106**. After removal and telescoping, the bicycle carrier **100** includes a minimum of five C-hoops **106**. Those of skill in the art will recognize in light of the present disclosure that other maximum or minimum numbers of C-hoops **106** are possible without departing from the scope of the present disclosure.

(96) FIG. 21 illustrates a C-hoop **106** and a portion of angled support members **108**, in accordance with an embodiment. The C-hoop **106** includes a straight portion **138** coupled to the angled support members **108** extending between a first bend **134** and a second bend **136**. The C-hoop **106** includes lengths **190** and **192** extending from the first and the second bends **134** and **136**. The lengths **190** and **192** extend outward. This outward extension helps to ensure that the spokes of a bicycle wheel will not contact the lengths **190** and **192** when the wheel is positioned in the C-hoop **106**. The lengths **190** and **192** may connect to each other. Alternatively, the lengths **190**, **192** can end in terminations **130**, **132** that define a gap **131**, as shown in FIGS. 12 and 13. In one embodiment, the lengths **190**, **192** extend outward at an angle between 5 degrees and 45 degrees. The C-hoop **106** can include single unitary bar or rod bent in the shape shown in FIG. 21. Alternatively, the C-hoop **106** can include multiple segments connected together.

(97) FIG. 22A is a view of a portion of a sports rack **2200** in raised position including damper **2204** configured as a force transmission **2206** with a force member **2208** for providing powered lifting or human assisted lifting capability, according to an embodiment. FIG. 22B is a view of the portion of the sports rack **2200** of FIG. 22A in the lowered position or intermediate position, according to an embodiment.

(98) Referring to FIGS. 22A and 22B, a sports rack **2200** for a vehicle **124** may include a hitch bar **112** configured to mount to a vehicle trailer hitch receiver **127**, a hinge assembly **114** operatively coupled to the hitch bar **112**, and at least one post **102** having a proximal end and a distal end. The at least one post **102** may be operatively coupled to the hinge assembly **114** at the proximal end, and the at least one post **102** may be rotatable relative to the hitch bar **112** through an arc in a vertical plane defined by the hinge assembly **114** between a lowered position and a raised position. The sports rack **2200** may include an equipment mount **103** operatively coupled to the distal end of the at least one post **102**, and a damper **152**, **2204** operatively coupled to the at least one post **102** and the hitch bar **112**. The damper **152**, **2204** may be configured to limit a rate of rotation of the at least one post **102** from the raised position to the lowered position.

(99) FIG. 23 is a view of a sports rack **100**, **2200** mounted on a vehicle **124**, the sports rack **100**,

2200 being shown in a raised position, ready for transport, according to an embodiment.

(100) Referring again to FIGS. **22A** and **22B**, in an embodiment, the damper **152, 2204** is configured to cause the at least one post **102** to rotate from the raised position to the lowered position at a rate slower than a rate of rotation driven by gravity. In an embodiment, the damper **152, 2204** includes a force transmission **2206** configured to cause the at least one post **102** and the equipment mount **103** to rotate from the lowered position to the raised position. In another embodiment, the damper **152, 2204** includes a force transmission **2206** configured to urge the at least one post **102** toward the raised position from the lowered position. The force transmission **2206** may be configured to exert sufficient torque to cause the at least one post **102** to rotate from the lowered position to the raised position without human lifting effort when the equipment mount **103** is empty. The force transmission **2206** may be configured to exert sufficient torque to reduce an amount of human lifting effort required or eliminate the human lifting effort altogether to rotate the at least one post **102** from the lowered position to the raised position when the equipment mount **103** is loaded to an equipment lifting capacity. In an embodiment, the force transmission **2206** includes a force multiplier, such as a gear set, a pulley, a cogged pulley, or a block and tackle, for multiplying a force received from a force member **2208**. The force transmission **2206** may include one or more of a lead screw mechanism, a rack and pinion mechanism, or a cogged belt and at least two cogged pulleys. The force transmission **2206** may include a crank for receiving a rotational force from a human user. Alternatively, the force transmission **2206** may include a coupling for receiving a rotational force from a drill or impact driver. In an embodiment, the force transmission **2206** includes a force member **2208** comprising a spring mechanism. In an embodiment, the force transmission **2206** includes a force member **2208** comprising a pneumatic cylinder. The force transmission **2206** may include a force member **2208** comprising a winch. Alternatively, the force transmission **2206** may include a force member **2208** comprising a brushless synchronous electric motor. The force transmission **2206** may include a force member **2208** configured to draw electrical current from a trailer wiring socket (not shown).

(101) In an embodiment, the sports rack **2200** further includes a locking mechanism **168** configured to stabilize the at least one post **102** in the raised position. In an embodiment, the sports rack **2200** further includes a detent mechanism or locking mechanism **168** configured to stabilize the at least one post **102** in the lowered position. At least one post **102** may consist essentially of one post **102**. In an embodiment, the at least one post **102** includes a formed post **102** including an off-axis portion disposed to provide a clear field to a vehicle **124** back-up camera (not shown) when the formed post **102** is in the raised position. In another embodiment, the at least one post **102** includes a formed post **102** including an off-axis portion disposed to provide for user access to a vehicle **124** rear hatch control (not shown) when the formed post **102** is in the raised position. Alternatively, the at least one post **102** may include two or more posts **102**, the two or more posts **102** disposed lateral to a centerline of the hitch bar **112**.

(102) In an embodiment, the equipment mount **103** includes a bicycle rack **100** configured to continuously hold one or more bicycles **126a, 126b** while the at least one post **102** is rotated from the lowered position to the raised position to support the one or more bicycles **126a, 126b** on the vehicle **124** above ground. The bicycle rack **100** may include a C-hoop **106** coupled to the first support member **104** and having a shape like a letter C.

(103) Referring to FIG. **23**, the equipment mount **103** may include a kayak rack configured to hold one or more respective kayaks **2300**. The equipment mount **103** may include a slot (not shown) configured to receive one or more kayak **2300** T-handles (not shown) disposed near a bow and/or stern (not shown) of respective kayaks **2300**. In an embodiment, the equipment mount **103** includes a mount **106** (such as a C-hoop, as shown) configured to hold either a bicycle wheel **128** or to hook a cockpit of a kayaks **2300**. In an embodiment, the one or more kayaks **2300** include whitewater kayaks **2300**. The one or more kayaks **2300** may be supported in a substantially vertical orientation when the at least one post **102** is in the raised position.

(104) In an embodiment, the bicycle carrier **100** may include integrated locking cables as an anti-theft deterrent when riders are not with the bicycles **126** and bicycle carrier **100**.

(105) Referring to FIGS. **22-23**, the sports rack **100, 2200** may further include a wall mount bar (see, e.g., wall mount structure **144** of FIG. **9A**) configured to be screwed to a wall (see, e.g., wall **145** of FIG. **9A**), and one or more wall mount fittings (see, e.g., wall mount fittings **149** of FIG. **9B**) configured to be coupled to the wall mount bar and configured to receive and hold a feature of the equipment mount **103**. In an embodiment, the sports rack **100, 2200** further includes a cable lock loop (see, e.g., cable lock loop **115** of FIG. **1**) coupled to the post **102** or equipment mount **103**. The cable lock loop may be configured to receive one or more locking cables or chains for securing sports equipment to the sports rack **100, 2200**.

(106) While various aspects and embodiments have been disclosed herein, other aspects and embodiments are contemplated. The various aspects and embodiments disclosed herein are for purposes of illustration and are not intended to be limiting, with the true scope and spirit being indicated by the following claims.

Claims

1. A bicycle carrier with a hitch mount, comprising: a hitch bar configured to mount to a vehicle trailer hitch receiver; a hinge assembly operatively coupled to the hitch bar, the hinge assembly including a damper operatively coupled to a support post and the hitch bar, the damper being configured to limit a rate of rotation of the support post relative to the hitch bar; the support post having a proximal end and a distal end, the support post being operatively coupled to the hinge assembly at the proximal end, the support post being rotatable relative to the hitch bar through an arc in a vertical plane defined by the hinge assembly between a lowered position and a raised position; a first support member operatively coupled to the distal end of the support post; and a wheel-receiving member coupled to the first support member, the wheel-receiving member being formed and configured to receive a wheel of a bicycle such that the bicycle is suspended vertically while the wheel is in the wheel-receiving member, the wheel-receiving member being further formed and configured to receive the wheel at any wheel angle within a continuous range of wheel angles with respect to the support post, the range of wheel angles including at least all angles between and including an angle substantially parallel to the support post and an acute angle with respect to the support post.
2. The bicycle carrier with the hitch mount of claim 1, further comprising a locking latch pin that, when engaged, is configured to prevent rotation of the support post.
3. The bicycle carrier with the hitch mount of claim 2, wherein the locking latch pin is configured to hold the support post in a plurality of selected positions including at least the lowered loading stop position and the raised transport position.
4. The bicycle carrier with the hitch mount of claim 3, wherein in the raised transport position, the support post is oriented vertically.
5. The bicycle carrier with the hitch mount of claim 4, wherein, when the support post is in the vertical position, the wheel receiving member defines a plane substantially parallel to the ground and perpendicular to the vertical direction.
6. The bicycle carrier with the hitch mount of claim 3, wherein in the lowered loading stop position, the support post is oriented at about 33° from vertical.
7. The bicycle carrier with the hitch mount of claim 2, further comprising a hinge release lever operatively coupled to the locking latch pin and configured to release the locking latch pin when pulled by a user.
8. The bicycle carrier with the hitch mount of claim 1, further comprising a hinge stop pin configured to define angles to which the support post is rotatable.
9. The bicycle carrier with the hitch mount of claim 1, wherein the damper includes a fluid

chamber and a piston, wherein fluid flow into or out of the fluid chamber is inhibited by an aperture.

10. The bicycle carrier with the hitch mount of claim 1, wherein the damper includes a spring.

11. The bicycle carrier with the hitch mount of claim 10, wherein the spring includes a gas spring.

12. The bicycle carrier with the hitch mount of claim 1, wherein the damper is pneumatic or hydraulic.

13. The bicycle carrier with the hitch mount of claim 1, wherein the wheel receiving member is configured to support the wheel by contact with a tire of the wheel without contacting spokes of the wheel or frame of the bicycle.

14. The bicycle carrier with the hitch mount of claim 13, wherein, after being positioned in a wheel receiving member, the wheel of the bicycle comes to rest at an angle other than vertical.

15. The bicycle carrier with the hitch mount of claim 14, wherein the wheel of the bicycle rests against an angled support member when the wheel of the bicycle comes to rest.

16. The bicycle carrier with the hitch mount of claim 1, wherein the wheel receiving member is configured to receive the wheel at a wheel angle substantially parallel to the support post and to then permit the wheel to rotate to a wheel angle of at least 20 degrees relative to the support post.

17. The bicycle carrier with the hitch mount of claim 1, wherein the wheel receiving member is about 26 inches long, perpendicular to the first support member.

18. The bicycle carrier with the hitch mount of claim 1, wherein the bicycle is suspended vertically while the wheel is in the wheel-receiving member when the support post is in the raised position.

19. The bicycle carrier with the hitch mount of claim 1, wherein the wheel receiving member is in the form of a "C".

20. The bicycle carrier with the hitch mount of claim 1, wherein the wheel-receiving member is coupled to the first support member and supported only by a coupling member disposed between the wheel-receiving member and the first support member, the coupling member extending from only one side of the wheel-receiving member.

21. The bicycle carrier with the hitch mount of claim 20, wherein the coupling member is angled relative to the first support member at an angle greater than 20 degrees from vertical when the support post is in the raised position.

22. The bicycle carrier with the hitch mount of claim 1, wherein the wheel receiving member includes a plurality of wheel receiving members configured to receive wheels of a corresponding plurality of bicycles.

23. The bicycle carrier with the hitch mount of claim 22, wherein the plurality of wheel receiving members includes seven wheel receiving members.

24. The bicycle carrier with the hitch mount of claim 22, wherein the plurality of wheel receiving members are spaced across the first support member to support the plurality of bicycles without mechanical interference between the plurality of bicycles.

25. A bicycle carrier with a hitch mount, comprising: a hitch bar configured to mount to a vehicle trailer hitch receiver; a hinge assembly operatively coupled to the hitch bar, the hinge assembly including a damper operatively coupled to a support post and the hitch bar, the damper being configured to limit a rate of rotation of the support post relative to the hitch bar; the support post having a proximal end and a distal end, the support post being operatively coupled to the hinge assembly at the proximal end, the support post being rotatable relative to the hitch bar through an arc in a vertical plane defined by the hinge assembly between a lowered position and a raised position; a first support member operatively coupled to the distal end of the support post; and a wheel-receiving member coupled to the first support member, the wheel-receiving member being configured to receive a wheel of a bicycle such that the bicycle is suspended vertically while the wheel is in the wheel-receiving member, the wheel-receiving member being further configured to receive the wheel at any of a range of angles with respect to the support post, the range of angles including at least an angle substantially parallel to the support post and an acute angle with respect

to the support post, wherein the damper is disposed in a retracted state while the support post is in the raised position.
